

Practical No.6: *Implement program to use clustering algorithms to find patterns in data

X. Conclusion

Clustering algorithms were successfully used to find hidden patterns in the dataset. K-Means grouped similar data points into clusters, and the results were visualized using graphs. Thus, clustering helps in understanding data and making better decisions.

XI. Practical Related Questions – Simple Answers

1. What is clustering? Difference between clustering and classification

Clustering is an **unsupervised learning** technique that groups similar data together.

Clustering	Classification
No labels in data	Data is labeled
Finds hidden groups	Predicts category
Example: customer groups	Example: spam detection

2. How to decide number of clusters (k)?

We use methods like:

- ✓ **Elbow Method** (most common)
- ✓ **Silhouette Score**

In Elbow method, we select **k** where **graph bends like an elbow**

3. Steps in K-Means Algorithm

1. Choose number of clusters (k)
 2. Select random centroids
 3. Assign each point to nearest centroid
 4. Recalculate centroids
 5. Repeat until centroids do not change
-

4. How to visualize clusters?

We use:

- ✓ **Matplotlib**
- ✓ **Seaborn**

Plots used:

- Scatter plot
- Color different clusters
- Show centroids

5. Difference: K-Means vs Hierarchical Clustering

K-Means	Hierarchical
Needs k value	No need of k initially
Fast	Slow
Uses centroids	Uses dendrogram tree
Flat clusters	Tree structure

6. Real-life applications of clustering

- ✓ Customer segmentation
 - ✓ Image compression
 - ✓ Recommendation systems
 - ✓ Disease pattern detection
 - ✓ Market analysis
-

7. Hard clustering vs Soft clustering

Hard Clustering	Soft Clustering
One data point → one cluster	One point → multiple clusters
Example: K-Means	Example: Fuzzy C-Means

XII. Exercise Answers

1. Write a program to use clustering algorithms to find patterns in data.

Dataset: Students grouped based on **Marks & Study Hours**

```
import pandas as pd
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Dataset (Students)
data = {
    'Marks': [35, 40, 42, 70, 75, 78, 90, 92, 95],
    'Hours': [2, 3, 2, 6, 7, 6, 9, 10, 9]
}

df = pd.DataFrame(data)

# Features
X = df[['Marks', 'Hours']]

# KMeans clustering
kmeans = KMeans(n_clusters=3, random_state=0)
df['Cluster'] = kmeans.fit_predict(X)

print("Student Groups based on performance:\n")
print(df)
```

Student Groups based on performance:

	Marks	Hours	Cluster
0	35	2	1
1	40	3	1
2	42	2	1
3	70	6	0
4	75	7	0
5	78	6	0
6	90	9	2
7	92	10	2
8	95	9	2

Meaning

- Cluster 0 → Low performers
- Cluster 1 → Average students
- Cluster 2 → Top performers

2. Write a program to implement K-Means Clustering algorithms

Dataset: Customers grouped by Spending & Visits

```
import pandas as pd
from sklearn.cluster import KMeans

# Dataset (Customers)
data = {
    'Spending': [2000, 2500, 3000, 8000, 8500, 9000, 15000, 16000, 17000],
    'Visits': [2, 3, 2, 7, 8, 7, 15, 16, 14]
}

df = pd.DataFrame(data)

# Features
X = df[['Spending', 'Visits']]

# Train model
kmeans = KMeans(n_clusters=3, random_state=0)
kmeans.fit(X)

# Results
df['Cluster'] = kmeans.labels_
centers = kmeans.cluster_centers_

print("Cluster Centers:\n", centers)
print("\nClustered Customer Data:\n", df)
```

```
Cluster Centers:
[[8.50000000e+03 7.33333333e+00]
 [1.60000000e+04 1.50000000e+01]
 [2.50000000e+03 2.33333333e+00]]
```

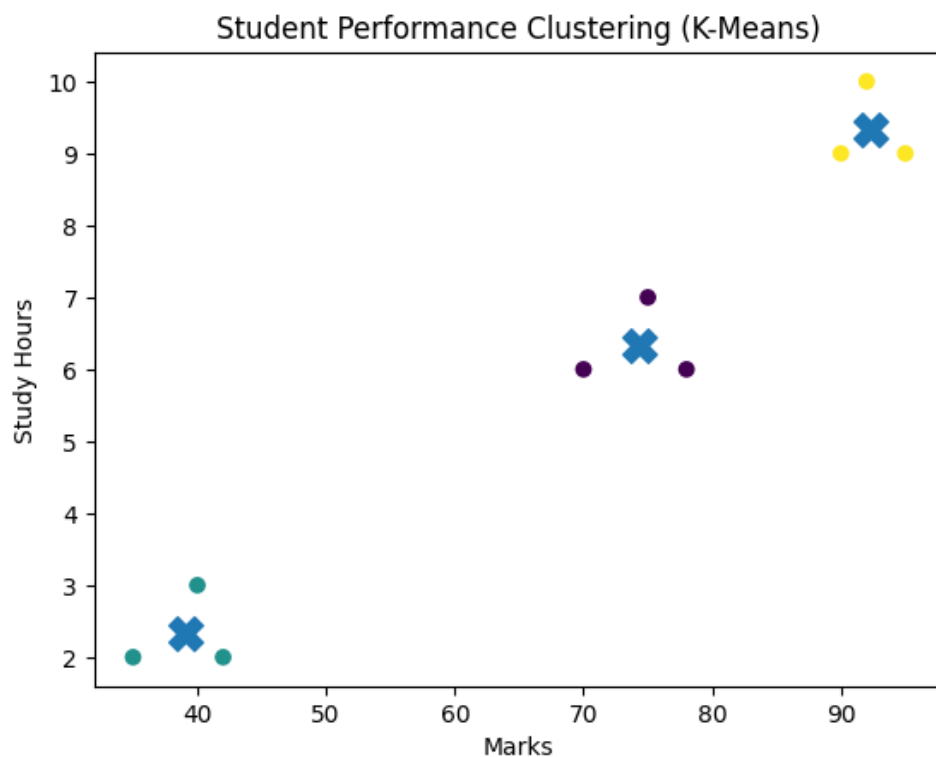
```
Clustered Customer Data:
   Spending  Visits  Cluster
0     2000      2         2
1     2500      3         2
2     3000      2         2
3     8000      7         0
4     8500      8         0
5     9000      7         0
6    15000     15         1
7    16000     16         1
8    17000     14         1
```

Meaning

- Cluster 1 → Low-value customers
- Cluster 2 → Medium customers
- Cluster 3 → Premium customers

3. Visualize the results using Matplotlib / Seaborn

```
: # ----- Visualization of Program 1-----  
plt.figure()  
  
# Scatter plot of students  
plt.scatter(df['Marks'], df['Hours'], c=df['Cluster'])  
  
# Plot centroids  
centroids = kmeans.cluster_centers_  
plt.scatter(centroids[:, 0], centroids[:, 1], marker='X', s=200)  
  
# Labels and title  
plt.xlabel("Marks")  
plt.ylabel("Study Hours")  
plt.title("Student Performance Clustering (K-Means)")  
  
plt.show()
```



Graph Description

- Each point = one student
- X-axis = Marks
- Y-axis = Study Hours
- Different colors = different student groups (clusters)
- "X" marks = cluster centers (centroids)

The graph shows how students are grouped based on **marks and study hours (performance level)**.

```

: # ----- Visualization of Program 2-----

import matplotlib.pyplot as plt

plt.figure(figsize=(7,5))

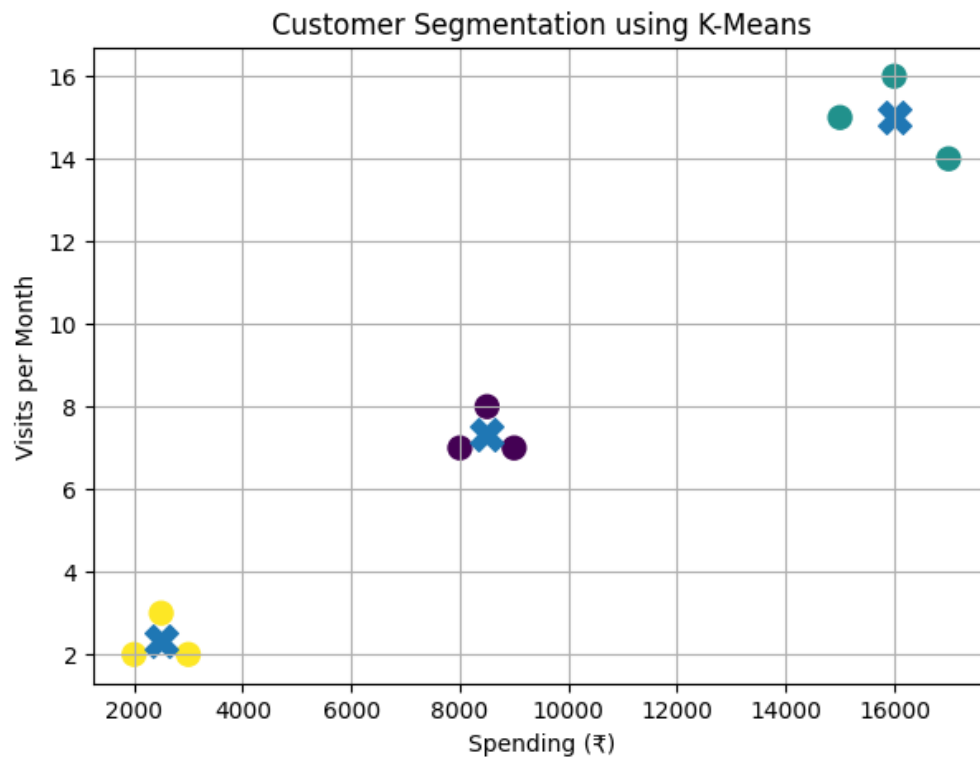
# Scatter plot
plt.scatter(df['Spending'], df['Visits'], c=df['Cluster'], s=100)

# Centroids
plt.scatter(centers[:,0], centers[:,1], marker='X', s=200)

plt.xlabel("Spending (₹)")
plt.ylabel("Visits per Month")
plt.title("Customer Segmentation using K-Means")
plt.grid(True)

plt.show()

```



Graph Description

- Each point = one customer
- X-axis = Age
- Y-axis = Spending Score
- Different colors = different customer groups (clusters)
- "X" marks = cluster centers (centroids)

The graph shows how customers are grouped based on **age and spending behaviour**.